

Seunghun Oh

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Education

- Seoul National University (SNU)** Mar 2020 – Aug 2025
BS in Mechanical Engineering, Double Major in Computer Science
- **GPA: 3.94/4.0, Major GPA: 3.97/4.0** Top 3.8% (8th of 207, Mechanical Engineering, SNU)
 - 21 months of military service included
- Gyeonggi Science High School for the Gifted (GSHS)** Mar 2017 - Feb 2020
Major in Physics and Mathematics
- 1 Year Early Entrance

Research Interests

- Robot Perception — localization and Mapping/SLAM, and 3D reconstruction
- Multi-Sensor Fusion — LiDAR–camera–IMU integration for robust perception
- Toward Autonomous Robotic Systems — interaction, decision-making, and reliable operation

Publications

- H. Song, D. Lee, **S. Oh**, M. Jung, and A. Kim*
 “The City That Never Settles: Simulation-based LiDAR Dataset for Long-term Place Recognition under Extreme Structural Changes”
Proceedings of the IEEE International Conference on Robotics and Automation (ICRA) Workshop, 2025.
Best Paper Award
[arXiv:2505.05076](#) [🔗](#)
- **S. Oh**[‡], Y. Kim[‡], C. Song, and A. Kim*
 “LiDAR Data Processing Algorithm for Robust 6-DoF Estimation Using Circular Patterns”
Journal of the Korean Robotics Society (KROS), June 2025.
[KROS Journal Link](#) [🔗](#)
- S. Hahn[‡], **S. Oh**[‡], M. Jung, A. Kim*, and S. Jung
 “Quantitative 3D Map Accuracy Evaluation Hardware and Algorithm for LiDAR (-Inertial) SLAM”
Proceedings of the International Conference on Control, Automation and Systems (ICCAS), 2024.
[arXiv:2408.09727](#) [🔗](#)

Research Experiences

- Undergraduate Researcher**, RPM Robotics Lab (Advisor: Prof. Ayoung Kim) Seoul, Korea
Seoul National University Dec 2023 – Aug 2025
- **3D Reconstruction with SDF-based Sparse Voxel Rasterization**
 Developed an SDF-based voxel rasterization pipeline to enhance mesh accuracy, extending SVRaster’s architecture. Focused on geometry refinement and enforcing voxel-wise continuity using CUDA and PyTorch.
 → [Open-source Project](#) [🔗](#)
 - **Long-term Place Recognition using Simulated LiDAR Datasets**
 Adapted and implemented long-term loop closure algorithms (Scan Context, BTC, Solid) in simulated urban environments. Evaluated PR performance under structural changes with customized pipelines.
 - **LiDAR–Camera Calibration via Circular Marker Detection**
 Developed the first robust LiDAR-based framework for accurate 6-DoF pose estimation using circular patterns, later integrated with the *DiscoCal (CVPR)* framework to complete a LiDAR–camera calibration system.
 → [Open-source Project](#) [🔗](#)
 - **Quantitative Evaluation of 3D SLAM Map Accuracy**
 Designed experimental setups and metrics for evaluating LiDAR(-inertial) SLAM accuracy. Developed benchmarking tools for 3D map reconstruction quality comparison. → [Open-source Project](#) [🔗](#)

Research Student, Aerospace Engineering Club (Bulnabi)
Seoul National University

Seoul, Korea
 Dec 2023 – Apr 2024

- Built foundational robotics environment using Ubuntu and ROS
- Simulated drone operations with Gazebo and QGroundControl

Research Student, Mechanical Engineering Club (ZERO)
Seoul National University

Seoul, Korea
Mar 2020 – Aug 2020

- Developed C++ and Linux programming skills for robotics platforms
- Gained early exposure to ROS-based development and embedded system configuration

Selected Awards and Honors

Scholarships / Fellowships

- **The National Presidential Science Scholarship**, Korea Student Aid Foundation 2024 – 2025
 Full tuition plus \$3700 each year for honorable undergraduates from the Korean government (4 semesters)
- **Work-Study Scholarship** (*Value Exploration and Practice*), Seoul National University Mar 2025 – Nov 2025
 Financial support of \$1500 for the work on the student society
- **Work-Study Scholarship** (Type 1), Seoul National University Mar 2024 – Nov 2024
 Financial support of \$2200 for the work on the student society
- **Merit-based Scholarship**, Dept. of Mechanical Engineering, Seoul National University Spring – Fall 2021

Awards and Honors

- 2025 – Co-authored ICRA 2025 workshop paper [1]; awarded **Best Research Award** (1st Place), IEEE ICRA Workshop on Future of Construction
- 2025 – **Outstanding Undergraduate Research Opportunities Program Award**, Dept. of Mechanical Engineering, Seoul National University
- 2024 – **Grand Prize**, Mechatronics Competition
- 2022 – **Winner**, Startup Camp hosted by Hanbat University (During Military Service)
- 2021 – Academic Honor: Highest GPA in 2nd Semester (4.3/4.3)
- 2018 – Selected for Winter School for National Team, Korean Physics Olympiad (KPhO)
- 2018 – **Awardee**, Korean High School Physics Olympiad (A0, A+)
- 2018 – **Winner**, SNU Youth Engineering Frontier Camp

Presentations / Talks

- Poster Presentation, “LiDAR Data Processing Algorithm for Robust 6-DoF Estimation Using Circular Patterns” Presented at KROS Spring Conference, Gangwon, Korea, 2025
- Seminar Talks, “Introduction to Linux”, “Drone software & simulation” “Introduction to ROS” [link](#) [🔗](#)
 Robotics Club Seminar, Seoul National University, 2024

Leadership / Extracurricular Activities

Republic of Korea Air Force (ROKAF)
Sergeant, 15th Air Force Base, 256th Squadron

Seoul, Korea
Mar. 2022 – Dec. 2023

- Served as squad leader, supervising 10 airmen in daily operations and training
- Managed Confidential Level II documents and coordinated flight schedules with pilots
- Ensured operational readiness and safety at the 256th Squadron, 15th Air Force Base

Mentoring Volunteer Program
Sports day and physics team leader

Andong, Korea
Jan. 2025 – Feb. 2025

- Led a team project organizing physics lessons and extracurricular science sessions for high school students
- Guided hands-on experiments and provided STEM mentoring for underserved students

Skills

Programming: C++, C, Java, Python, MATLAB, RISC-V Assembly, Bluespec

Libraries/Softwares: Pytorch, CUDA, ROS1-2, Arduino, Fusion360, L^AT_EX